

Linux Commands Cheat Sheet for Cloud Engineering

1. Navigation Commands

Command	Meaning
<code>pwd</code>	Show current directory
<code>ls</code>	List files
<code>ls -la</code>	List all files with permissions
<code>cd foldername</code>	Go into a folder
<code>cd ..</code>	Go back one directory
<code>cd ~</code>	Go to home directory
<code>clear</code>	Clear terminal screen

```
pwd
ls
ls -la
cd /var/log
cd ..
cd ~
```

clear

2. File and Directory Commands

Command	Meaning
<code>touch file.txt</code>	Create empty file
<code>mkdir folder</code>	Create folder
<code>mkdir -p a/b/c</code>	Create nested folders
<code>cp file1 file2</code>	Copy file
<code>cp -r folder1 folder2</code>	Copy folder
<code>mv old new</code>	Rename or move
<code>rm file.txt</code>	Delete file
<code>rm -r folder</code>	Delete folder
<code>cat file.txt</code>	Show file content
<code>less file.txt</code>	View large file page by page
<code>head file.txt</code>	Show first 10 lines
<code>tail file.txt</code>	Show last 10 lines
<code>tail -f file.txt</code>	Watch file live

```
touch notes.txt
mkdir logs
cp notes.txt backup.txt
mv backup.txt old-notes.txt
rm old-notes.txt
cat notes.txt
less /var/log/syslog
tail -f /var/log/syslog
```

3. Search and Text Processing

These are very important for log analysis.

Command	Meaning
<code>grep "word" file</code>	Search word in file
<code>grep -i "word" file</code>	Case-insensitive search
<code>grep -r "word" folder</code>	Recursive search
<code>find /path -name file</code>	Find file by name
<code>wc -l file</code>	Count lines
<code>sort file</code>	Sort lines

<code>uniq file</code>	Remove duplicate lines
<code>cut -d ":" -f1 file</code>	Extract column
<code>awk '{print \$1}' file</code>	Print column
<code>sed 's/old/new/g' file</code>	Replace text

```
grep "Failed password" /var/log/auth.log
grep -i "error" /var/log/syslog
find /etc -name "*.conf"
wc -l /var/log/auth.log
cat access.log | awk '{print $1}' | sort | uniq -c
```

4. Permissions and Ownership

This is very important for cloud security.

Command	Meaning
<code>ls -l</code>	View permissions
<code>chmod 755 file</code>	Change permission

<code>chmod +x script.sh</code>	Make script executable
<code>chown user:group file</code>	Change owner
<code>whoami</code>	Show current user
<code>id</code>	Show user/group IDs
<code>groups</code>	Show user groups
<code>sudo command</code>	Run as admin

Permission numbers:

Number	Meaning
7	read + write + execute
6	read + write
5	read + execute
4	read only
0	no permission

Common examples:

```
ls -l
```

```
chmod 644 file.txt
```

```
chmod 755 script.sh
```

```
chmod +x security-audit.sh
```

```
sudo chown ubuntu:ubuntu file.txt
```

```
whoami
```

```
id
```

```
groups
```

Example permission meaning:

```
-rwxr-xr-x
```

Means:

- Owner: read/write/execute
- Group: read/execute
- Others: read/execute

5. User and Group Management

Useful for Linux security hardening.

Command	Meaning
<code>sudo adduser username</code>	Create user

<code>sudo userdel username</code>	Delete user
<code>sudo usermod -aG group user</code>	Add user to group
<code>sudo passwd username</code>	Change password
<code>cat /etc/passwd</code>	View users
<code>cat /etc/group</code>	View groups
<code>sudo groupadd groupname</code>	Create group

```
sudo adduser cloudops
sudo groupadd securityteam
sudo usermod -aG securityteam cloudops
sudo passwd cloudops
cat /etc/passwd
cat /etc/group
```

6. Process Management

Cloud engineers troubleshoot running processes all the time.

Command	Meaning
<code>ps aux</code>	Show running processes

<code>top</code>	Live process monitor
<code>htop</code>	Better live monitor
<code>kill PID</code>	Stop process
<code>kill -9 PID</code>	Force stop process
<code>pgrep name</code>	Find process ID
<code>pkill name</code>	Kill process by name

```
ps aux
top
htop
pgrep nginx
sudo kill 1234
sudo pkill nginx
```

7. System Resource Monitoring

Used for server health checks.

Command	Meaning
<code>uptime</code>	Show system uptime/load

<code>free -h</code>	Show memory usage
<code>df -h</code>	Show disk usage
<code>du -sh folder</code>	Folder size
<code>lscpu</code>	CPU info
<code>lsblk</code>	Disk/block devices
<code>uname -a</code>	Kernel/system info

uptime
 free -h
 df -h
 du -sh /var/log
 lscpu
 lsblk
 uname -a

8. Service Management with systemd

Very important for cloud servers.

Command	Meaning
<code>systemctl status service</code>	Check service

<code>systemctl start service</code>	Start service
<code>systemctl stop service</code>	Stop service
<code>systemctl restart service</code>	Restart service
<code>systemctl enable service</code>	Start on boot
<code>systemctl disable service</code>	Disable on boot
<code>journalctl -u service</code>	View service logs

```
sudo systemctl status nginx
sudo systemctl start nginx
sudo systemctl stop nginx
sudo systemctl restart nginx
sudo systemctl enable nginx
sudo journalctl -u nginx
sudo journalctl -u nginx -f
```

9. Networking Commands

These are extremely important for cloud engineering.

Command	Meaning
<code>ip addr</code>	Show IP addresses
<code>ip route</code>	Show routing table
<code>ping domain.com</code>	Test connectivity
<code>curl URL</code>	Test HTTP/API
<code>wget URL</code>	Download file
<code>ss -tulnp</code>	Show open ports
<code>netstat -tulnp</code>	Older open-port command
<code>dig domain.com</code>	DNS lookup
<code>nslookup domain.com</code>	DNS lookup
<code>traceroute domain.com</code>	Show route path

`ip addr`
`ip route`
`ping google.com`
`curl http://localhost`
`curl -I https://example.com`
`ss -tulnp`
`dig google.com`
`nslookup google.com`
`traceroute google.com`

Open port checking:

```
sudo ss -tulnp
```

Common ports:

Port	Service
22	SSH
80	HTTP
443	HTTPS
3306	MySQL
5432	PostgreSQL
6379	Redis
8080	Web app/dev server

10. Firewall Commands

Ubuntu usually uses `ufw`.

Command	Meaning
---------	---------

<code>sudo ufw status</code>	Check firewall
<code>sudo ufw enable</code>	Enable firewall
<code>sudo ufw disable</code>	Disable firewall
<code>sudo ufw allow 22</code>	Allow SSH
<code>sudo ufw allow 80</code>	Allow HTTP
<code>sudo ufw deny 3306</code>	Deny MySQL
<code>sudo ufw delete allow 80</code>	Remove rule

```
sudo ufw status
sudo ufw enable
sudo ufw allow OpenSSH
sudo ufw allow 80
sudo ufw allow 443
sudo ufw deny 3306
sudo ufw status numbered
```

For cloud security, this connects directly to:

- Security Groups
- NACLs
- Firewall rules
- Network segmentation

11. Package Management

Ubuntu/Debian:

```
sudo apt update
sudo apt upgrade
sudo apt install nginx
sudo apt remove nginx
apt search docker
apt list --installed
```

Amazon Linux / RHEL / CentOS:

```
sudo yum update
sudo yum install nginx
sudo yum remove nginx
```

Newer RHEL/Fedora:

```
sudo dnf update
sudo dnf install nginx
```

12. SSH and Remote Server Commands

Very important for EC2 and cloud servers.

Command	Meaning
---------	---------

<code>ssh user@ip</code>	Connect to server
<code>scp file user@ip:/path</code>	Copy file to server
<code>rsync -av folder user@ip:/path</code>	Sync folder
<code>ssh-keygen</code>	Generate SSH key

```
ssh ubuntu@SERVER_IP
ssh -i key.pem ubuntu@SERVER_IP
scp file.txt ubuntu@SERVER_IP:/home/ubuntu/
rsync -av project/ ubuntu@SERVER_IP:/home/ubuntu/project/
ssh-keygen
```

Fix key permission issue:

```
chmod 400 key.pem
```

13. Log Analysis Commands

Very useful for security and troubleshooting.

Common log files:

```
/var/log/auth.log
```

```
/var/log/syslog
```

```
/var/log/nginx/access.log
/var/log/nginx/error.log
/var/log/cloud-init.log
```

Useful commands:

```
sudo tail -f /var/log/auth.log
sudo grep "Failed password" /var/log/auth.log
sudo grep "sudo" /var/log/auth.log
sudo tail -f /var/log/nginx/access.log
sudo tail -f /var/log/nginx/error.log
sudo journalctl -xe
```

Count failed SSH attempts:

```
sudo grep "Failed password" /var/log/auth.log | wc -l
```

Find suspicious IPs:

```
sudo grep "Failed password" /var/log/auth.log | awk '{print $(NF-3)}' |
sort | uniq -c | sort -nr
```

14. Archive, Backup, and Compression

Command	Meaning
<code>tar -czvf file.tar.gz folder</code>	Compress folder

<code>tar -xzvf file.tar.gz</code>	Extract archive
<code>zip -r file.zip folder</code>	Zip folder
<code>unzip file.zip</code>	Unzip
<code>rsync -av source dest</code>	Backup/sync

```
tar -czvf logs-backup.tar.gz /var/log
tar -xzvf logs-backup.tar.gz
zip -r project.zip project/
unzip project.zip
rsync -av /var/log/ ./backup-logs/
```

15. Disk and Storage Commands

Command	Meaning
<code>df -h</code>	Disk space
<code>du -sh folder</code>	Folder size
<code>lsblk</code>	Block devices
<code>mount</code>	Show mounted filesystems
<code>blkid</code>	Show disk UUIDs

df -h
du -sh /var/log
lsblk
mount
sudo blkid

Cloud connection:

- EC2 EBS volumes
- Disk full issues
- Log rotation
- Backup systems

16. Environment Variables

Command	Meaning
<code>env</code>	Show environment variables
<code>echo \$PATH</code>	Show PATH
<code>export</code> <code>VAR=value</code>	Set variable
<code>printenv</code>	Print env variables

env
printenv

```
echo $PATH
export APP_ENV=production
echo $APP_ENV
```

17. Bash Scripting Basics

Create script:

```
nano health-check.sh
```

Script example:

```
#!/bin/bash

echo "System Health Report"
echo "===== "
echo "Uptime:"
uptime

echo "Memory:"
free -h

echo "Disk:"
df -h

echo "Open Ports:"
ss -tulnp
```

Make executable:

```
chmod +x health-check.sh
```

Run:

```
./health-check.sh
```

Important Bash syntax:

```
#!/bin/bash
```

```
name="Waliur"  
echo "Hello $name"
```

```
if [ -f "file.txt" ]; then  
    echo "File exists"  
else  
    echo "File missing"  
fi
```

```
for file in *.log; do  
    echo "Checking $file"  
done
```

18. Cron Jobs

Used for scheduled automation.

Open cron editor:

```
crontab -e
```

Run script every day at 2 AM:

```
0 2 * * * /home/ubuntu/health-check.sh
```

Cron format:

```
* * * * * command
```

```
| | | | |
```

```
| | | | day of week
```

```
| | | month
```

```
| | day of month
```

```
| hour
```

```
minute
```

Examples:

```
* / 5 * * * * command    # every 5 minutes
```

```
0 * * * * command        # every hour
```

```
0 2 * * * command        # every day at 2 AM
```

```
0 2 * * 0 command        # every Sunday at 2 AM
```

Cloud connection:

- EventBridge cron
 - Scheduled Lambda
 - Backup automation
 - Log rotation
-

19. Docker-Related Linux Commands

Since Docker is important for cloud:

```
docker --version
docker ps
docker ps -a
docker images
docker build -t app .
docker run -p 8080:80 app
docker stop CONTAINER_ID
docker rm CONTAINER_ID
docker rmi IMAGE_ID
docker logs CONTAINER_ID
docker exec -it CONTAINER_ID bash
```

20. Security Audit Commands

Check logged-in users:

```
who
w
last
```

Check sudo users:

```
getent group sudo
```

Check failed logins:

```
sudo grep "Failed password" /var/log/auth.log
```

Check open ports:

```
sudo ss -tulnp
```

Check running services:

```
systemctl list-units --type=service --state=running
```

Check file permissions:

```
find / -perm -4000 2>/dev/null
```

Find world-writable files:

```
find / -type f -perm -0002 2>/dev/null
```

Commands You Should Memorize First

For your cloud/security path, memorize these first:

```
pwd
```

```
ls -la
```

```
cd
```

```
cat
```

less
head
tail -f
grep
find
chmod
chown
sudo
ps aux
top
kill
systemctl
journalctl
ip addr
ip route
ping
curl
ss -tulnp
dig
df -h
free -h
ssh
scp
rsync
tar
cron

Most Important Cloud-Relevant Linux Skills

If you master only these 7 areas, you will already be ahead of many CS students:

1. File permissions
2. Users and groups
3. Service management
4. Log analysis
5. Networking troubleshooting
6. Bash scripting
7. SSH/server administration

Next, I can give you the **step-by-step procedure to build the Linux CloudOps Command Center project** with folders, scripts, README sections, screenshots checklist, and resume bullet points.